

HENAN, China

WMO: 560650

Lat: 34.7343N

Lon: 101.6011E

Elev: 3501

StdP: 65.76

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-23.6	-31.7	0.3	-18.4	-29.7	0.4	-15.9	10.9	-0.5	9.7	-2.0	0.3	90	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	20.4	11.8	18.9	10.8	17.4	10.0	13.1	18.7	12.2	17.2	11.4	15.8	2.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	12.7	16.1	10.3	12.1	14.9	9.6	11.5	14.0	50.0	18.8	47.0	17.3	44.3	15.8	16.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	7.5	6.3	-30.2	23.2	2.3	1.1	-31.9	24.0	-33.3	24.7	-34.6	25.3	-36.3	26.1
			-30.5	14.8	2.3	0.7	-32.1	15.3	-33.5	15.8	-34.7	16.2	-36.4	16.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.0	-11.3	-8.0	-3.1	2.4	5.8	9.0	11.2	10.9	7.6	1.9	-4.8	-10.2
	DBStd	8.51	3.76	4.03	3.47	3.06	2.54	2.50	2.39	2.63	3.03	3.09	3.57	3.28
	HDD10.0	3417	661	505	405	229	131	47	14	20	83	251	444	627
	HDD18.3	6324	919	738	663	479	387	279	220	229	321	509	694	886
	CDD10.0	134	0	0	0	0	2	18	52	49	13	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	2.2	1.9	2.2	2.6	2.7	2.6	2.4	2.2	2.0	2.1	1.9	1.8	1.8
Precipitation	PrecAvg	597	4	6	13	28	72	96	128	110	96	38	6	2
	PrecMax	888	17	16	24	72	154	176	259	261	184	113	18	6
	PrecMin	452	0	0	1	7	40	41	52	42	46	8	0	0
	PrecStd	94	4	4	7	14	23	29	46	48	36	23	4	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	9.5	13.4	17.0	18.3	20.1	22.7	22.1	20.6	15.5	9.7	5.8
		MCWB	-2.9	-0.9	1.0	4.0	6.4	11.2	12.8	13.0	12.0	6.1	0.1	-2.7
	2%	DB	2.7	6.0	9.7	14.0	16.0	17.9	20.6	20.2	17.8	12.4	7.1	2.9
		MCWB	-4.5	-2.7	-0.4	2.9	5.9	9.4	11.9	11.9	10.1	4.4	-1.2	-4.3
	5%	DB	0.2	3.6	7.3	11.9	14.1	16.4	18.9	18.6	15.5	10.1	5.3	0.7
		MCWB	-5.9	-3.8	-1.4	2.0	5.2	8.8	11.1	11.2	8.9	3.0	-2.0	-5.6
	10%	DB	-2.2	1.3	5.1	9.8	12.4	14.9	17.2	17.0	13.6	8.2	3.4	-1.4
		MCWB	-7.3	-4.9	-2.4	1.4	4.5	8.2	10.5	10.4	7.8	2.2	-2.9	-6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	-0.8	1.9	5.9	9.1	13.1	14.2	14.3	13.3	8.2	0.7	-2.3
		MCDB	5.2	8.5	11.1	13.7	14.7	18.1	19.8	20.4	18.6	13.0	8.0	5.2
	2%	WB	-4.3	-2.3	0.2	4.1	7.7	11.5	13.1	13.1	11.4	6.0	-0.5	-4.1
		MCDB	2.4	5.0	7.9	11.4	13.2	15.8	18.8	18.6	16.3	10.1	5.7	2.4
	5%	WB	-5.8	-3.5	-0.9	3.0	6.5	10.2	12.2	12.2	10.1	4.4	-1.5	-5.4
		MCDB	0.0	3.0	6.0	9.5	11.6	14.3	17.3	17.1	14.3	8.3	4.1	0.4
	10%	WB	-7.2	-4.7	-1.9	2.1	5.5	9.2	11.3	11.4	8.9	3.1	-2.6	-6.6
		MCDB	-2.4	0.8	4.2	8.3	10.5	13.0	15.7	15.6	12.2	6.8	2.6	-1.5
Mean Daily Temperature Range	5% DB	MDBR	18.6	17.2	14.5	12.6	11.6	10.1	11.1	11.4	11.2	12.1	15.9	18.5
		MCDBR	22.0	21.0	19.1	17.2	15.7	13.5	15.0	14.5	14.6	15.5	18.4	21.3
		MCWBR	15.4	13.4	10.2	7.5	6.9	6.2	7.0	6.9	7.3	8.0	10.7	14.7
	5% WB	MCDBR	21.0	19.5	16.5	13.8	11.8	10.3	11.6	11.3	11.7	11.9	16.3	20.3
		MCWBR	14.8	12.7	8.9	6.3	6.0	5.2	5.8	5.5	6.1	6.7	9.7	14.2
Clear-Sky Solar Irradiance	taub	0.220	0.256	0.308	0.343	0.340	0.331	0.322	0.317	0.292	0.265	0.229	0.207	
	taud	2.519	2.421	2.249	2.181	2.293	2.412	2.477	2.480	2.539	2.553	2.564	2.585	
	Ebn at Noon	1020	1003	964	940	943	946	953	956	973	982	1000	1024	
	Edh at Noon	79	99	129	146	132	117	109	107	96	86	75	69	
All-Sky Solar Radiation	RadAvg	3.37	4.16	5.16	5.90	5.77	5.51	5.73	5.39	4.47	4.04	3.63	3.25	
	RadStd	0.20	0.21	0.25	0.29	0.24	0.46	0.56	0.47	0.40	0.29	0.12	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.54	+0.91	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.42	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page